Smart City

Prototype - Team Gold

Weston Carmack, Carlos Moll Rolon, Nick Kooshki, Tony Cairuz, Derrick Stewart, Joshua Connelly

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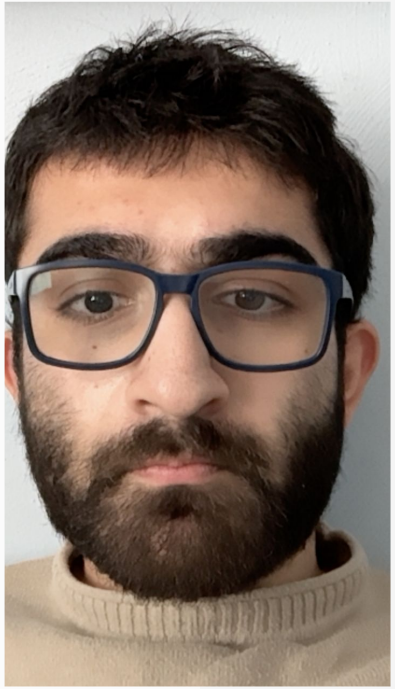
Meet the team

Tony Cairuz

I'm passionate about sports, especially skiing, and I was proud to become the champion of skiing in Lebanon for a while. Outside of sports, I'm studying Computer Science and working alongside my studies.

(Maps & Navigation, Frontend)



Nick Kooshki

I am currently a senior at ODU with a major in computer science. My biggest interests in technology are computer hardware alongside reviewing MacBooks and exploring the latest artificial intelligence trends and capabilities.

(Forum & Integration)

Meet the team

Carlos Moll Rolon

I am currently a senior at ODU studying Computer Science. I enjoy being able to problem-solve through code. Some hobbies of mine outside of school are lifting weights, listening to music, reading manga, and cooking.

(Authentication & Database, Backend)

Derrick Stewart

I'm a senior at ODU studying Computer Science. I enjoy technology, music, and working on creative projects.

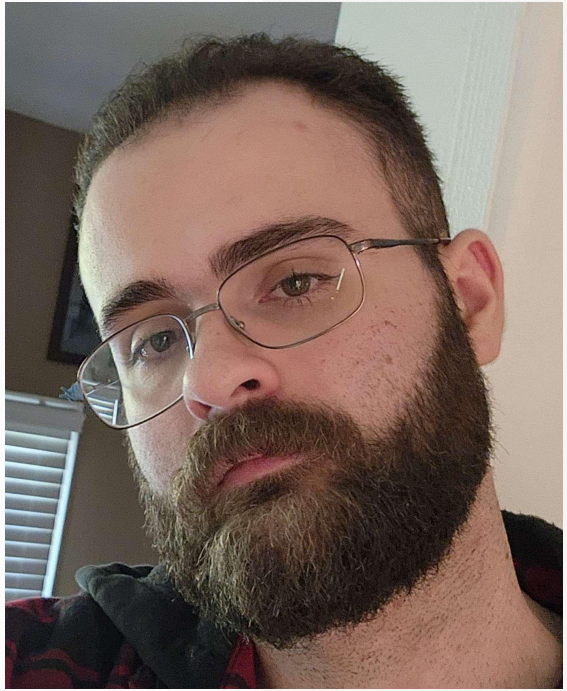
(Forum & Integration)

Meet the team

Weston Carmack

I'm currently a senior in computer science here at ODU. My favorite programming languages are C++ and Python, and plan on picking up Rust in the future. Outside of class I enjoy watching football, Formula 1, working out and attending car shows.

(Team Lead, Website, Authentication & Database, Backend)



Joshua Connelly

I am a senior studying computer science with a preference for coding. I have the most experience in Java, but my favorite is C#. My goal is to be a programmer and code my own game in my off-time.

(Maps & Navigation, Frontend)

Problem Statement

- **What's happening?**
  - Rapid urbanization and poor planning create overcrowding, reducing quality of life.
- **Why It's Happening**
  - Mass migration to urban areas
  - Data gaps leading to poor urban planning causing:
    - Health Issues
    - Traffic Jams
    - Poor Quality Housing
    - Inadequate Waste Management
    - Water Scarcity
    - Increased Crime Rate
- **Impact**
  - Longer commutes, unaffordable housing, and reduced community well-being.

Sprint 1

Project & TDD Setup

- Setup GitHub repository
- Setup React Native project
- Database Setup
- Maps API Setup

Final Touches Before Sprint 2

- Optimize database to best ability
- Finish API endpoints
- Build login UI Screens
- Test Integrations
- Start adding:
  - Map controls
  - Forum controls

Week 1

Week 2

Week 3

API & Component Foundation

- Setup database operation
- Setup API endpoints
- Start getting UI libraries in order
- Start Maps integration from Mapbox
- Start basic forum page setup

Sprint 2

Authentication Implementation

- Make adjustments to database for user authentication
- Testing Authentication
- Connect Login Screens

Week 5

Week 4

Auth Polish & Integration

- Setup database session management for users
- Test maps authentication
- Test forum authentication
- Polish UI authentication

Sprint 3

Search Functionality

- Verify location database schema
- Work on search implementation w/ Mapbox
- Create map markers when search is made
- Create search UI for forum
- Make test for the forum search functionality

Implementing More Advance Features

- Work on geospatial queries for “posts near me”.
- Work on map performance optimization
- Test all features (halfway point)
- Implement offline maps
- Add various map styles

Week 6 — **Week 7** — **Week 8**

Basic Routing

- Setup routing storage
- Work on the route rendering and overview
- Adjust the navigation controls
- Test various routing locations, edge cases, etc.

Sprint 4

Forum & Community

- Verify the forum schema
- Create post UI & Test post creation
- Work on “create post at this location”
- Refine the forum feed
- Implement forum feed onto the map

Week 10

Week 9

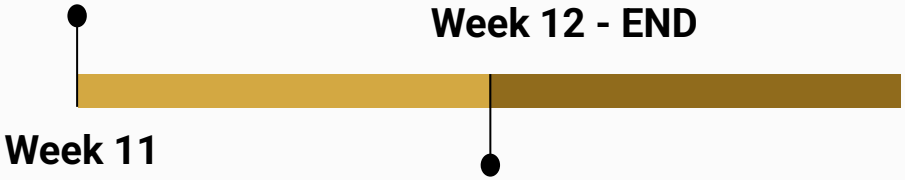
Comments and Interactions

- Verify the Comments schema
- Ensure forum security for only logged in users
- Build post detail UI
- Implement the comments functionality

Sprint 5

Systems Improvements

- Analyze the database performance and make improvements
- Setup more API error handling
- Verify the UI Consistency across all pages
- Check the cross-feature performance and integration
- Test the user flow for logging in and signing up
- Build loading screens



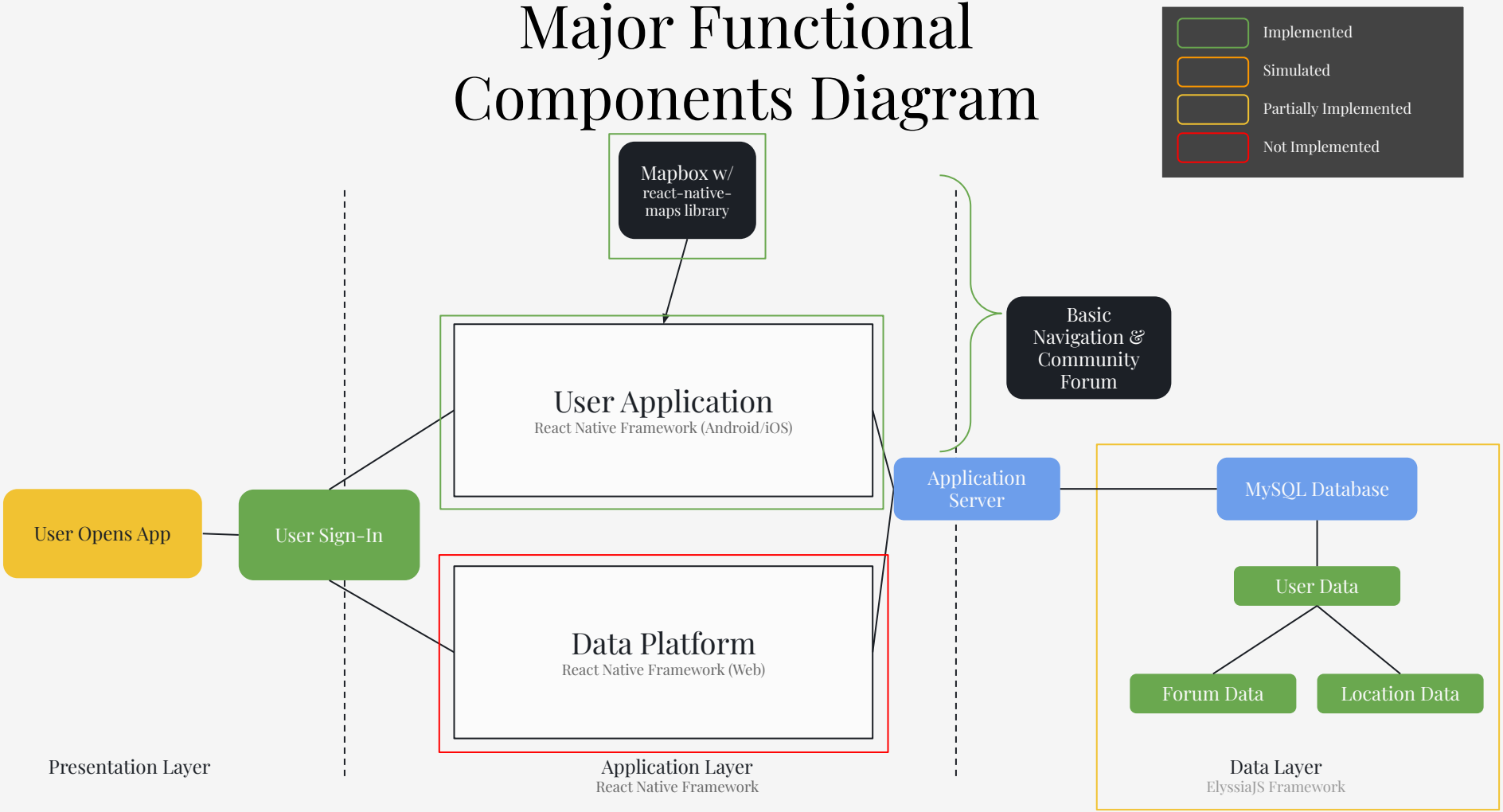
Final Preparations

- Demo Database, API features, UI, Scenarios and different environments
- Fix any bugs and perform final testing
- Prep for final presentation
- Finish up documentation and any other loose ends

Development Tools

Platform		
Component	Language(s)	JavaScript & TypeScript
	Framework	React Native + Android Studio
	IDE	Visual Studio Code
	Backend	ElysiaJS
	Database	MySQL
	Version Control	Git
	Issue Tracking	GitHub
	APIs	Mapbox + OpenStreetMap
	Testing	Bun

Major Functional Components Diagram



Feature Table: RWP vs. Prototype

Category	Features	RWP	Current Prototype	Reasoning
Account Management	Login/Authentication	Fully Implemented	Fully Implemented	Essential for prototype
	Account Creation/Deletion	Fully Implemented	Fully Implemented	Essential for prototype
	Manage Account Access/Permissions	Fully Implemented	Fully Implemented	Essential for prototype
User Application	Access Community Forum	Fully Implemented	Fully Implemented	Essential for prototype
	Access Dynamic Queue System for Navigation	Fully Implemented	Partially Implemented	We removed the Dynamic Queue implementation as it would be too complex for the given time frame. Still implementing the basic navigation.
	Alerts and Notifications	Fully Implemented	Partially Implemented	Simulated feature until we figure out how to handle API calls.
	Add Parking locations	Fully Implemented	Not Implemented	No time given scope of project. Could reconsider

Feature Table: RWP vs. Prototype Continued

Category	Features	RWP	Current Prototype	Reasoning
Data Platform	Central Information Hub	Fully Implemented	Partially Implemented	Still will implement the database for storing information in. Will drop long-term data platform due to time constraints.
	Data Analytics and Insights	Fully Implemented	Not Implemented	Dropped due to no longer implementing the long-term data platform
	Visualize Data	Fully Implemented	Not Implemented	Dropped due to no longer implementing the long-term data platform
	Data Anonymity	Fully Implemented	Fully Implemented	Essential for prototype

User Stories

1. As a resident, I want to securely log into my account so that I can access city services and community features.
2. As a resident, I want to create an account to save my info.
3. As a resident, I want to delete my account so that I can remove my personal information from the system if I want.
4. As a resident, I want to access the community forum so that I can create and view post and communicate with other residents.
5. As a resident, I want to view navigation info so that I can move around the city.
6. As a resident, I want to receive alerts or notifications about posts that I may find important.
7. As a resident, I want my personal data to be anonymized so that my privacy is protected when using the platform.

User Stories to Features

User Story	Feature
As a resident, I want to securely log into my account so that I can access city services and community features.	Login/Authentication
As a resident, I want to create an account to save my info.	Account Creation
As a resident, I want to delete my account so that I can remove my personal information from the system if I want.	Account Deletion

User Stories to Features Cont'd

User Story	Feature
As a resident, I want to access the community forum so that I can create and view post and communicate with other residents.	Access Community Forum
As a resident, I want to view navigation info so that I can move around the city.	Basic Navigation System (Simplified from Dynamic Queue)
As a resident, I want to receive alerts or notifications about posts that I may find important.	Alerts and Notifications
As a resident, I want my personal data to be anonymized so that my privacy is protected when using the platform.	Data Anonymity

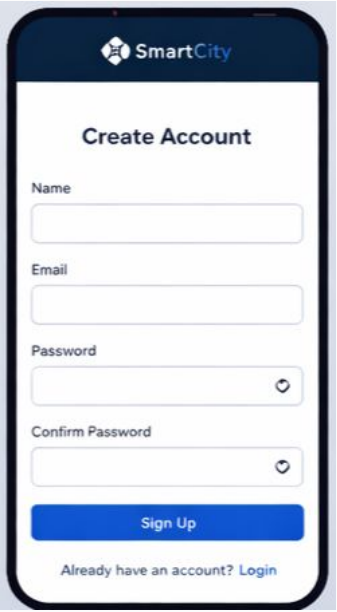
User Stories to Tasks

User Story	Tasks
As a resident, I want to securely log into my account so that I can access city services and community features.	Implement the user authentication system, including the login interface and validation.
As a resident, I want to create an account to save my info.	Develop the user registration functionality, including account creation and storage of user information.
As a resident, I want to delete my account so that I can remove my personal information from the system if I want.	Implement account management functionality to allow users to delete their accounts and data.

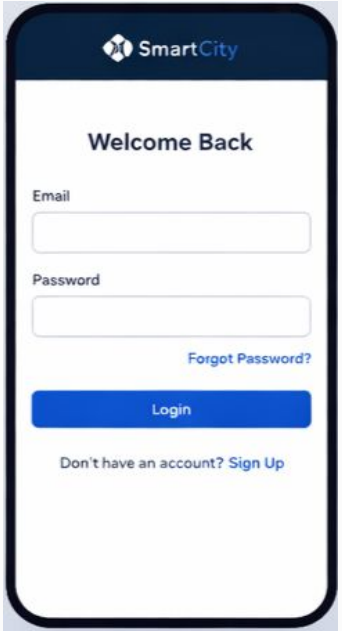
User Stories to Tasks Cont'd

User Story	Tasks
As a resident, I want to access the community forum so that I can create and view post and communicate with other residents.	Implement the community forum feature, including the forum interface and displaying posts.
As a resident, I want to view navigation info so that I can move around the city.	Develop the navigation feature, including map integration
As a resident, I want to receive alerts or notifications about posts that I may find important.	Implement the alerts and notifications feature
As a resident, I want my personal data to be anonymized so that my privacy is protected when using the platform.	Implement data privacy and anonymization for user information

UI Mockup

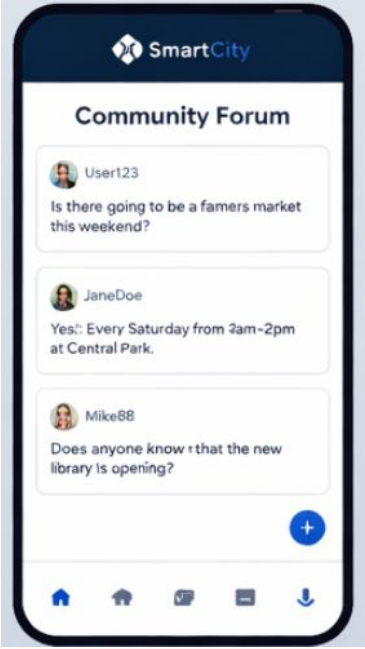


Account Creation



Login

Homepage/Map



Forum

Customer Risks

Risk Matrix			Impact				
			Very Low	Low	Medium	High	Very High
			1	2	3	4	5
Probability	Very High	5					
	High	4				C3R	
	Medium	3		C2M	C1R, C3M	C2R	
	Low	2			C1M		
	Very Low	1					

Security Risks

Risk Matrix			Impact				
			Very Low	Low	Medium	High	Very High
			1	2	3	4	5
Probability	Very High	5		S2R			
	High	4					
	Medium	3		S2M S1R			
	Low	2	S1M				
	Very Low	1					

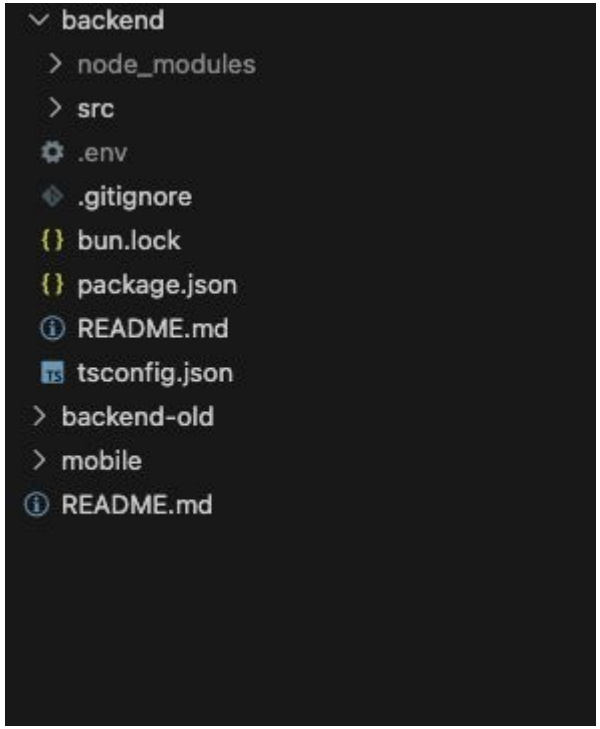
Technical Risks

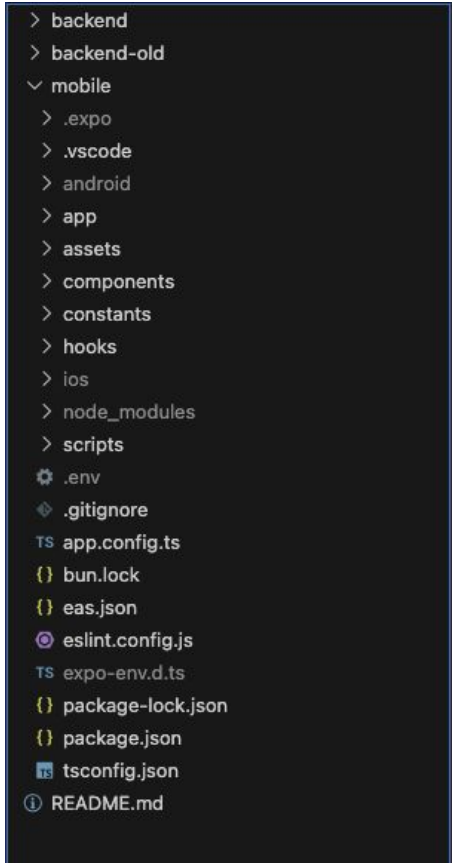
Risk Matrix			Impact				
			Very Low	Low	Medium	High	Very High
			1	2	3	4	5
Probability	Very High	5					
	High	4					T4R
	Medium	3				T3R T4M	
	Low	2			T3M	T1R	
	Very Low	1			T1M		

Issues/Challenges

- **Learning Curve:** Learning JavaScript and TypeScript along with the some development tools.
- **Frontend & Backend Integration**: Navigating the learning curve of securely connecting our React Native (JavaScript) frontend to the ElysiaJS backend and routing API calls to the database.
- **Database & VM Deployment**: Setting up our MySQL database in Coder Workspace hosted on the CS Department VM.

Repository Setup Status





Next Steps: Carlos Moll Rolon

Broad Tasks:

- Setup connection between frontend and backend.
- Setup connection for forum posts.

Goal:

- Implement routes and controllers for connectivity between frontend and backend.
- Test with functions to display data on local host.

Next Steps: Joshua Connelly

Broad Tasks:

- Implement more map tools (Search, Routing, Improved Map Visualization, etc.)
- Implement a turn by turn Navigation UI.

Goal:

- Transform the GPS from a static map viewer into an interactive navigation tool.

Next Steps: Derrick Stewart

Broad Tasks:

- Complete implementation of map navigation
- Begin development of the community forum

Goal:

- Have the map fully functional
- Have a skeleton of the community forum

Next Steps: Nick Kooshki

Broad Tasks:

- Set up the basic community forum page and UI components
- Integrate forum controls and navigation into the main application

Goal:

- Deliver a functional initial community forum interface

Next Steps: Tony Cairuz

Broad Tasks:

- Improve UI flow between the map and other tabs
- Implement map search functionality with location update integration

Goal:

- Establish a seamless and responsive UI workflow across the application
- Ensure map search updates location dynamically and functions smoothly

Next Steps: Weston Carmack

Broad Tasks:

- Setup Coder Workspace
- Get application routed to MySQL database
- Write MySQL queries revolved around account creation and verification

Goal:

- Get account creation and verification setup in the backend
- Build workflow for further implementation of other modules from the client

References (Part 1)

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Demo Showcase